

Emanuel Sanchez

Chicago, IL | emanuels8001@gmail.com | 2244333192
Linkedin: [emanuel-sanchez](#) | Github: <https://github.com/emanuels8>

Summary:

Full-Stack Software Engineer with 4+ years of experience building scalable web applications using TypeScript, React.js, Go, Ruby. Proficient in developing responsive, modern UIs with Tailwind CSS, and building robust backend systems with RESTful and GraphQL APIs. Experienced in AWS cloud services (Lambda, S3, EC2, RDS). Skilled in designing real-time, event-driven architectures using Kafka and WebSockets. Familiar with integrating LLMs and AI-powered features into applications. Strong focus on clean, maintainable code and end-to-end product delivery.

Skills:

Backend & Infrastructure: Go, Ruby, Kafka, Redis, Kubernetes, AWS, Nodejs (Cognito, Lambda, DynamoDB, EKS)
Frontend: React.js, TypeScript, JavaScript, CI/CD (AWS, GitHub Actions)

Professional Experience:

TCWGlobal - (Contracted to GitHub)

Remote

Software Engineer - Customer Success Team

Sep 2025 - Current

- Collaborated cross-functionally with product, support, and engineering teams to deliver customer-focused solutions at scale.
- Developed and maintained full-stack features using Ruby, TypeScript, and React to improve usability and reliability of customer-facing workflows.
- Built internal tools and services in Go to improve operational efficiency and reduce manual effort for support teams.
- Contributed to updates and ongoing maintenance of GitHub's Customer Support platform (support.github.com), with a focus on stability, performance, and user experience.
- Participated in code reviews and continuous improvement efforts to ensure maintainable, high-quality production systems.

DriftNet

Chicago, IL

Software Engineer - Team Lead

Aug 2021 - Feb 2025

- Architected scalable microservices using Node.js and Prisma ORM, supporting 100K+ monthly API requests with PostgreSQL and DynamoDB as primary data stores.
- Deployed and maintained serverless APIs on AWS Lambda behind API Gateway, achieving 99.99% uptime.
- Utilized Lambda layers and environment specific configurations to ensure modular, reusable, and secure deployments across dev, staging, and production.
- Implemented dynamic routing, authentication (e.g., NextAuth), and state management using React Context, Redux, and RTK Query in Next.js applications.
- Developed a virtual assistant feature using Ollama for natural language navigation, allowing users to interactively explore the product website through chat. Integrated LLM prompts with page metadata and user context to enable accurate routing, significantly improving user engagement and support experience.
- Managed cloud storage workflows with AWS S3, handling 500K+ media files using pre-signed URLs, automatic archival policies, and cross-region replication.
- Built rich client-side dashboards using React and Next.js to visualize real-time metrics and system status.
- Implemented dynamic filtering, WebSocket updates, and bundle size optimization for fast rendering on constrained networks.
- Built a dynamic breadcrumb feature in Next.js to guide first-time users through key sections of the product website. Contextually highlighted user journey steps based on route structure and interaction history, improving onboarding clarity.
- Secured authentication with AWS Cognito, MFA, social sign-ins (OAuth2), and custom role-based access control using JWTs.
- Designed DynamoDB table schemas and indexes to store and query large volumes of event data for a time-sensitive application, and configured partition and sort keys for the data, which resulted in improved query performance and reduced costs
- Developed event-driven Lambda functions for scheduling and delivering user-created event information via Amazon SNS, ensuring timely communication.

Education

College Of Lake County

Grayslake, IL

Computer Science

September 2016 - December 2019

- **Smart Parking System:** Developed a real-time web app using React.js and Firebase to track available parking spots on campus; implemented map integration and sensor data simulation.
- **Library Management System:** Built a CRUD-based system using Node.js, Express, and MongoDB to manage books, borrowers, and transactions; implemented role-based access control for admins and users.
- **Weather Dashboard:** Created a responsive weather app using JavaScript, OpenWeatherMap API, and Tailwind CSS to fetch and display real-time weather data with location search.
- **Course Scheduler App:** Designed and implemented a scheduling tool in Python with a GUI using Tkinter, allowing students to generate optimized class schedules based on course availability.
- **Sorting Algorithm Visualizer:** Developed a web-based visualization tool in TypeScript and React to demonstrate different sorting algorithms (Bubble, Merge, Quick) in real time.